Alejandro Jaimes

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science; Minor in Robotics — *Expected Grad: May 2028*

Experience

Software Engineer Intern — Aug 2026 - Present
BNY — *Lake Mary, FL*

- Improved service availability by **20%** via **Azure VM** monitor alerts and **Grafana** dashboards across **5** services.
- Engineered containerized **Docker** deployment workflow cutting provisioning time by **3 minutes** for **4+** engineers.

Software Engineer Intern — May 2026 - Present
Institute for Simulation and Training — *Orlando, FL*

- Deployed a shared spatial-anchor **XR** system at an ONR tech demo, keeping **10+** concurrent users aligned in one physical space across **15** hours and **300+** participants.
- Streamed anchor-relative transforms via **Photon RPC** at **10Hz**, enabling real-time co-located multi-user tracks.
- Constructed re-localization system to correct drift every **2s**, sustaining accuracy of user movement across **50m**.
- Built a **Firebase RTDB** pipeline that shares saved anchors across headsets automatically, eliminating per-device manual setup and saving an estimated **20+ hours** of setup time.

President — Apr 2026 - Present
Graphics Programming Knights — *Orlando, FL*

- Founded a non-profit student organization, scaling membership to **200+** within **11 months** of launch.
- Conducted **5+ workshops** on **GPU programming**, **mathematics**, and **game engines** with **70+** attendees.
- Led a team of **10+** engineers to maintain an **open-source** monorepository, actively used by **100+** members.
- Established partnerships with **5+** organizations to have events, leading to a **75%** membership increase.

Projects

Permanence | *C++, CUDA, Jetson Orin Nano, C#, Unity, Python, PyTorch*

- Placed **Best Overall of 212** projects, **1st of 21** for NVIDIA and **1st of 78** for Hugging Face at Knight Hacks IX.
- Wrote a fused **W8A16 matrix-vector CUDA kernel** in **C++/CUDA** (warp-per-row, 16-byte vectorized int8 loads, **shared-memory** activation staging) targeting **Jetson Orin Nano (sm_87)**.
- Built a **Unity/C#** AR HUD for **XREAL** glasses streaming live **Jetson** detections over **SSE** at up to **10 Hz**.

greghouse | *C++, Arduino, ESP32, React, AWS, Node.js, Lambda, DynamoDB*

- Awarded **3rd** in the **Best Overall** category at BloomKnights out of **86** projects and **240+** participants.
- Implemented **ESP32** firmware in **C++** streaming soil telemetry to **AWS** every **2s** with **sub-1s latency.**
- Designed IoT pipeline with **API Gateway**, **Lambda**, and **DynamoDB** handling **1,800+** readings **hourly**.

HIP Renderer | *C++, HIP, ROCm, Python, Linux*

- Developed a **GPU-accelerated** path tracer in **HIP** achieving **8.8x** speedup over CPU on **AMD MI300X**.
- Profiled kernel performance with **rocprof**, improving bandwidth by **55%** via **shared memory optimization**.
- Automated **GPU profiling** pipeline in **Python** analyzing **5** hardware metrics per optimization cycle on **Linux**.

sFun (SNES Emulator) | *C++, CMake, JSON, SDL2*

- Architected a **65C816 CPU** emulator in **C++** with 8/16-bit width switching, and 24-bit addressing.
- Designed a 256-entry function pointer dispatch table across 9 logical translation units, enabling O(1) opcode dispatch and improving codebase maintainability at scale.
- Engineered a **JSON-driven harness** with register-level diffing, achieving **100%** pass rate across **25+** test cases.

Technical Skills

Languages: C, C++, Assembly, Python, Java, C#, JavaScript, TypeScript, HTML, CSS
GPU & Systems: HIP, CUDA, ROCm, rocprof, OpenGL, CMake
Frameworks: Unreal Engine, Unity, Next.js, Flask, Node.js, OpenCV, Mediapipe, Three.js
Libraries & Tools: React, Tailwind CSS, SDL2, WebSockets, TFT_eSPI, Arduino, Docker, AWS, Git, Linux, ESP32, PlatformIO, Cloudflare, Vite, Photon RPC